ITA0501 - Computer Vision CO1 - AT2 - Concept Mapping M. SHAKTHI - 192312538

Autonomous Vehicle : Computer Vision Concept Map

Overview of Computer vision :
* Enables machines to understand visual information from the world.
* Uses cameras and algorithms to interpret images and make decisions.
* Combines image processing, pattern recognition and AI/ML.

Features of Computer Vision
* Image acquisition and processing
* Feature extraction
* object recognition and classification
* Real time processing
* Robustness to varying conditions

Levels of Computer vision and Image processing
* Low level - image enhancement, noise reduction
* Mid level - feature extraction, segmentation
* High level - object recognition, scene understanding

Digital Image Fundamentals
* Image represented as a 2D array of pixels
* Pixel values indicate intensity / color
* Grayscal or color images.
* Resolution and bit depth affect quality.

Computer Vision Pipelines

Camera Sensors (in car)
capture real-time images of the surroundings

Image Capture → Pre-Processing → Feature extraction → Object recognition → Navigation & Decision-making → **Output / Support**
Detects roads, pedestrians, bicycles, traffic signs, obstacles and supports safe navigation.

Image Sensing and acquisition
* CMOS / CCD cameras
* Lenses and optics
* Light conditions and exposure.
* Frame rate and resolution.

Sampling and Quantization
* Sampling : converting continuous image to discrete grid (pixels)
* Quantization : assigning intensity levels (eg : 8-bit → 256 levels)

Image formation models
* Pinhole camera model
* Perspective projection
* Lens Distortion
* Illumination and
* reflectance.

Applications
* Surveillance
* Autonomous vehicles
* Traffic monitoring
* Smart transportation Systems
* Robotics and drones

Medical Imaging : X-Ray Concept Map

Overview of Medical Imaging
- uses imaging techniques to visualize internal structures of the body.
- Helps doctors detect abnormalities, diseases and injuries.
- Provides non-invasive and detailed diagnostic information

Digital Image fundamentals
- Image as a 2D pixel array.
- Grayscale (X-Ray) or color.
- Pixel intensity represents tissue density.
- Resolution and bit depth affect clarity.

Image Sensing and acquisition
- X-ray source & tube
- Patient and detector
- X-ray photons pass through body
- Detector captures the signal

Sampling and Quantisation
- Sampling : Convert continuous signal to discrete pixels
- Quantisation : map to intensity levels (eg: 12-16 bit for medical images)

Image formation Models
- X-ray attenuation model.
- Beer-Lambert law
- Tissue density and contrast
- Projection geometry

X-Ray Imaging System
Captures X-ray, converts to digital image and sends for processing

Imaging Pipeline

X-Ray Capture → Digitization → Image Processing → Feature Extraction & analysis → Diagnosis support

Levels of Computer vision and image processing
- Low level - image enhancement, denoising.
- Mid level - segmentation, feature extraction
- High level - image interpretation, diagnosis

Features of Computer vision
- Improved image quality.
- Accurate feature detection
- Automated analysis
- Real time / efficient processing

Applications in medical imaging
- X-ray (chest, bone, dental)
- CT scan
- MRI
- ultrasound
- Disease detection and monitoring

Output / Diagnosis
Processed X-ray image helps doctors identify abnormalities and make accurate diagnosis.